

INFO 492 — INTENSIVE CAPSTONE

AI-Assisted RFI Review

Checking Information Before Submission

Team: Team 8 — Subcontractor Perspective

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Research Background: Making RFIs Clearer

- Unclear questions and missing context can lead to requests for more information.
- Procore recommends clear questions and relevant background information.
- Existing Procore AI tools help draft RFIs and find related project information.
- Our research asks whether an AI check before submission can reduce the GC's first-review time.

Footer citations: Medina & Tell (2025); Procore (n.d.).

Our Thesis: Check Before Submission

Proposed target: Reduce median GC first-review time by at least **25%**, without increasing information errors.

- Compare the same RFI form completed with and without AI assistance.
- The agent checks missing context, unclear questions, and document references.
- The subcontractor verifies the suggestions and approves the final submission.

Our Dataset: Dett Elementary School (Chicago)

A new one-story annex and renovation of an existing school, finished in 2024

A public project, so all six addenda with RFIs and answers are online.

Real questions from contractors and real answers from the architect.

Each addendum shows which drawings were changed, which gives us a correct answer for testing.

Preparing the Data

- 01** Each RFI becomes one row in a CSV file with the question, answer, drawings, and problem type.

- 02** We also build a list of drawing numbers, so the agent can check if a reference exists.

- 03** These RFIs are from the bid stage, so we rewrite them as subcontractor-to-GC RFIs.

- 04** Every rewritten RFI is labeled as simulated and linked to the original.

Four Demo Concepts

- 01 Find missing details:** remove information like location or deadline, and test if the agent notices.

- 02 Check references:** test if the agent catches wrong drawing references, such as a keynote that pointed to detail 6/A-703 instead of 1/A-703.

- 03 Improve the draft:** the agent makes unclear questions more specific, and the subcontractor approves the final version.

- 04 Send to the GC:** the GC team reviews RFIs with and without AI, and we record review time, errors, and clarification requests.

Coordination Posture: Parties

Party	Role in our study
Subcontractor (Sub Team 8)	Writes the RFI, reviews the agent's suggestions, and approves the final version before it goes to the GC.
AI agent	Flags missing context, unclear questions, and references not in the drawing index, with a source for each. Never submits, approves design changes, authorizes work, or commits cost or time.
General contractor (GC Team 1)	Does the first review and decides: forward to the architect, or return for more information. We time this decision.
Design team (Architect)	Answers RFIs. On Dett, the architect's real answers and addendum drawing changes are our correct answers for testing.
Owner (Public Building Commission of Chicago)	Not active in our experiments. Involved only if an RFI answer leads to a cost or time change through a change order.

Coordination Posture: Artifacts

IN SCOPE

- RFIs: The artifact we create, check, and send to the GC.
- Drawings & addenda: What the agent checks references against.
- Drawing-number index: Built from the Dett addenda so the agent can confirm a reference exists.

OUT OF SCOPE

- Submittals: Separate approval workflow from RFIs.
- Change orders: The agent may note a possible impact, but never creates or prices one.
- Daily reports & inspection records: Not available in the Dett public documents.